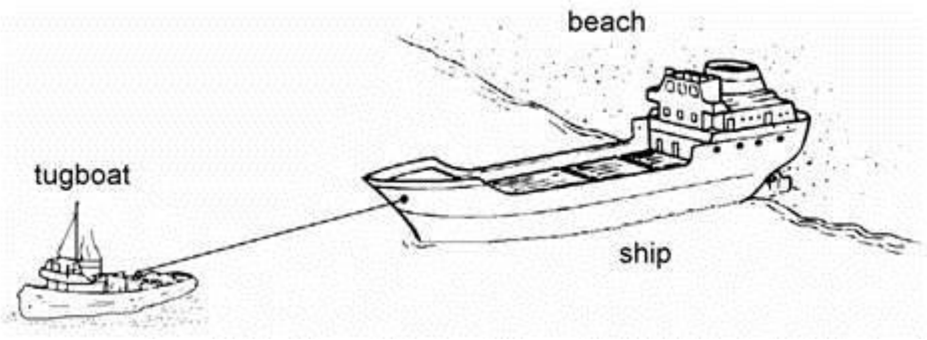
Q1. In a storm, a small ship was blown onto a beach. Now it is calm and there is no wind. A tugboat is trying to pull the ship off the beach.



(a) The tugboat pulls the ship with a force of 25 000 N.

The ship does not move because of the force of friction acting on it.

(i) Tick **one** box to show the size of the frictional force acting on the ship.

zero ☐

more than zero but less than 25 000 N ☐

25 000 N ☐

more than 25 000 N ☐

1 mark

(ii) Add an arrow to the drawing to show the direction of the frictional force acting on the ship.

1 mark

(b) When the tide is higher, the tugboat again pulls the ship with a steady force of 25 000 N. The ship begins to move.

Once the ship is off the beach, the tugboat continues to pull the ship with a force of 25 000 N.
A frictional force due to the water acts on the ship.

(i) At first, the speed of the ship increases.

Tick **one** box to describe the frictional force acting on the ship while its speed is increasing.

zero ☐

more than zero but less than 25 000 N ☐

25 000 N ☐

more than 25 000 N ☐

1 mark

(ii) After a short while, the ship reaches a steady speed. The tugboat continues to pull with a force of 25 000 N.

Tick **one** box to describe the frictional force acting on the ship while it is going at a steady speed.

zero ☐

more than zero but less than 25 000 N ☐

25 000 N ☐

more than 25 000 N ☐

1 mark

(iii) The ship is towed to the north. What is the direction of the frictional force acting on the ship?

....

1 mark
Maximum 5 marks

Mark schemes

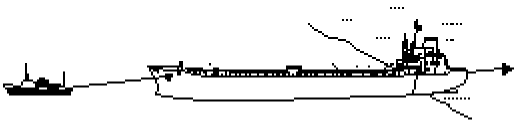
Q1.

(a) (i) 25 000 N √

if more than one box is ticked, award no mark

1 (L5)

(ii) an arrow pointing in the opposite direction to pull of the tow-rope as shown below



the arrow parallel to the tugboat may be drawn anywhere on the diagram

*do **not** accept a horizontal arrow*

1 (L5)

(b) (i) more than zero but less than 25 000 N

if more than one box is ticked award no mark

1 (L6)

(ii) 25 000N ✓

if more than one box is ticked, award no mark

1 (L6)

(iii) any **one** from

- south **or** southwards
- in the opposite direction to the movement **or** pull
- away from tugboat

*accept 'opposing motion' **or** 'backwards'*

1 (L5)

[5]